

Lee 05 Multinomial logit / probit

1. Multinomial logit / probit

↳ any individual characteristic

2. conditional logit

↳ individual & alternative charact.

→ No interpretation of parameter just sign

↳ increasing probability w respect to base

ME: $\beta \cdot 100$ p.p increase in probability

→ for all alternative
IIA → Hausman McFadden test. → independence of irrelevant alternatives

↳ if rejected → multinomial probit

R^2 → can't interpret

conditional ME

→ for all alternatives → for all alternative features.

Lee 04 ordered logit / probit

→ parameters just sign for higher / lower

↳ Lipsitz test Specification test.
 H_0 : the form of the model is appropriate

• Hosmer Lemeshow

H_0 : form is fine

• Pukstaus & Robinson → if we have dummy

• Brant → proportional odds assumption

H_0 : OK, 1st line

* we can't interpret McFadden R^2 statistics

ME

$\beta \cdot 100$ P.P.

What are the assumptions.

Lee 03

logit / probit → binary dep.

→ parameters → just sign → we can compare the strength if it is LOGIT

ME

$\beta \cdot 100$ P.P. → ↑ in probability of success

count

adj. → because of variation of explanatory variables
90% of all samples are correctly predicted.

McKelvey, Zavoina.

• If the latent variable is observed, our model would explain 70 variation.

• Likelihood → specification

• OLS - Rojek → H_0 : correct specification

• Ramsey RESET → specification / form
• bptest

Lee 02 panel data.

→ parameter → unit by unit.

• PFTest → Poolability OLS / panel

↳ BIASED & INCONSISTENT PARAMETER

↳ Autocorrelation & Heteroscedasticity.

H_0 : all individual effects are jointly OLS insignificant

→ If there is autocorrelation / Hetero. we can't trust std. err, t-value, p-value

st. errors → using robust estimator. ↑ errors

FE - If we suspect correlation b/w individual effects and Regressors

Random effects assumptions: Independence b/w individual effects and Regressors

• Hausman test

H_0 : RE is appropriate

H_1 : FE

TIP: $LP_{test} = 2 \left| \log_{\text{general}} - \log_{\text{restricted}} \right| \rightarrow \text{test statistic}$

$\chi^2_{q \rightarrow \text{NO. of Restriction}} \rightarrow \text{critical value}$



Granger causality

$$y_t = \alpha_0 + \sum_{i=1}^p \alpha_i y_{t-i} + \sum_{i=1}^q \beta_i x_{t-i}$$

H_0 : all lags of x are jointly insignificant
lags of $x \rightarrow$ doesn't cause y

Assumption: Only for stationary variables or cointegrated variables

R^2 within → a model explains % time variation in y
RE bin → " " " " across entities

Overall - the model explain % of overall variation of dep. variables.

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3. Hausman Test for endo. OLS
IV/OLS H_0 : Exogeneity. $cov(X, e) = 0$
 H_1 : Endogeneity IV

1. weak instrument $\rightarrow$ IV worse than OLS
 $F > 10$ or $F > 3.3 \rightarrow$ strong instrument
 $\rightarrow$ use p-value (more accurate.)

2. Instrument Validity (SARGAN TEST)
 H_0 : Instruments are OK

NOTE: R^2 can be -ve when we have endogeneity

Lee 13 co-integration

- $\rightarrow$ Regress $\rightarrow$ calculate Residuals
 $\rightarrow$ Stationarity of Residuals ADF "nc"
not p-value critical value

- $\rightarrow$ ECM $\rightarrow$ check significance.
 $\Delta y_t = \beta_0 + \beta_1 \Delta x_t + \alpha(Les) + \epsilon \rightarrow$ pg test
 $[1, -\hat{\beta}_0, -\hat{\beta}_1] \rightarrow$ long term run slip

* If not cointegrated consider $\rightarrow$ DL/ARDL

Lee 12 DL/ARDL $\rightarrow$ Retention

- $\rightarrow$ all assumptions of OLS has to be met

$$\begin{aligned} STM &= \beta_0 \\ LTM &= \frac{\sum \beta_0}{n} \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{DL}$$

$$LTM = \frac{\sum \beta_i}{1 - \sum \alpha_i} \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{ARDL}$$

- ARDL:
- stationarity $\rightarrow$ more v
- linear reg $\rightarrow$ lag
- Diagnostics $\rightarrow$ Ramsey $\rightarrow$ lag
Hama $\rightarrow$ Breusch Godfrey $\rightarrow$ lag
 $\rightarrow$ Breusch Pagan $\rightarrow$ lag
 $\rightarrow$ Normality.

ARMA/ARIMA AR

- $\rightarrow$ stationarity, select $(p, q) \rightarrow$ depending on frequency of data
 $\rightarrow$ Diagnostics $\rightarrow$ autocorrelation $\rightarrow$ Ljung Box
 $\rightarrow$ For short term prediction $\rightarrow$ up to $p+q$
 $\rightarrow$ No interpretation of parameter.

- DL } $\rightarrow$ check endogeneity
ARDL } $\rightarrow$ autocorrelation
 $\rightarrow$ Heteroscedasticity
 $\rightarrow$ specification
 $\rightarrow$ stationarity
 $\rightarrow$ Model fit

stationarity

ADF

PPSS

PP

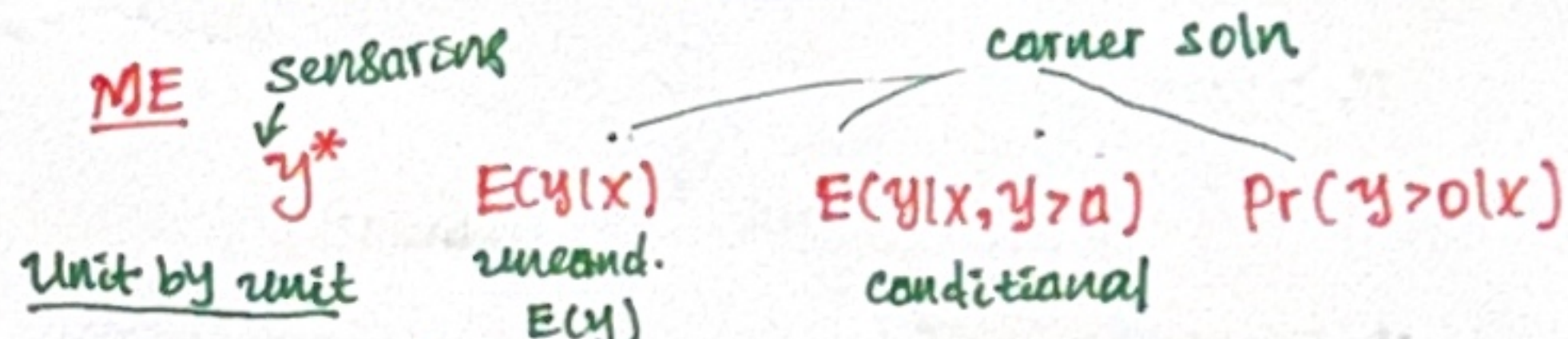
1. H_0 : random walk $\rightarrow$ name "nc"
 H_1 : st, $\mu = 0$ drift "a"
2. H_0 : random walk $\rightarrow$ drift
 H_1 : st, $\mu \neq 0$ trend "ct"
3. H_0 : Random walk $\rightarrow$ quadratic trend
 H_1 : trend stationarity



* ADF & PP $\rightarrow$ low power against $I(0)$ alternative that are close to $I(1)$

Lee 7 Tobit $\rightarrow$ cont. LDV

- $\rightarrow$ If no sig amount of obs. on the edge we can use OLS
corner soln $\rightarrow$ economic optimization (decision of entities)
* we can't interpret parameters just significance



Truncated Regression

- $\rightarrow$ unit by unit
 $\rightarrow$ we can interpret parameters

Lee 6 count data

- $y \in \{0, 1, \dots\}$ non-negative Number
 $\rightarrow$ we interpret parameters, no ME
 $\beta_i \rightarrow$ not p.p.t $\rightarrow$ Expected No. of count $\uparrow/\downarrow$ by β_i percent

- * Zero Inflated model
success = being in always zero group.

- * Hurdle success = being in non-zero group.

- # $\rightarrow$ LR test $() \Rightarrow$ H_0 poisson Reg vs. H_1 Neg. bin. Reg.
check over/under dispersion

- # Score test $\Rightarrow$ For excess zero
 H_0 : NO excess zero

1st block] count block

2nd block] inflation block